

Video Effects Selection Profile

Proposed additions for the video_effects-registry

Prepared for: Max (maxzzzzzzzzzzzzzzzzzzzz)

Purpose: Give Max a clear set of original effect concepts to review and select for implementation.

Scope: Six high-value effects identified from current mobile-editing workflows in CapCut, Alight Motion, Movavi, Adobe Premiere Rush, and Adobe Premiere on iPhone.

Decision requested: Please choose the effect or effects you would like implemented first. The proposed names are original registry names rather than direct copies of the names used by other editing applications.

Executive recommendation

The six candidates below cover the most useful visual roles in a modern mobile editor: cinematic lighting, abstract stylization, motion realism, impact energy, digital distortion, and dynamic transitions. All six can be expressed as effect entries with a dedicated `effect.json`, GLSL shader, preview image, and `registry.json` entry. Some candidates may benefit from application-side animation or multi-pass rendering, which is noted in the technical sections.

Priority	Proposed ID	Display name	Primary role	Implementation difficulty	Recommendation
1	prism_bloom	Prism Bloom	Cinematic light bloom and glow	Medium	Best first effect
2	mirror_orbit	Mirror Orbit	Radial symmetry and psychedelic stylization	Medium	Best distinctive effect
3	velocity_veil	Velocity Veil	Motion softness and energetic movement	Medium–High	Best motion effect
4	impact_tremor	Impact Tremor	Controlled camera shake for beats and impacts	Low–Medium	Fastest dynamic effect
5	neon_trace	Neon Trace	Glowing outlines and edge emphasis	Medium	Strong stylized effect
6	warp_surge	Warp Surge	Zoom streaks and push-through transitions	Medium–High	Best transition effect

1. Prism Bloom

Concept

Prism Bloom makes bright parts of a video appear to emit light. Lamps, windows, reflections, flames, bright text, and highlights develop a soft, colored halo while darker areas remain

comparatively stable. The concept is based on thresholded glow treatments documented in Alight Motion and the glow/light-leak effect family available in Adobe Premiere on iPhone.[1](#)

Exact function

The effect first measures the luminance of each pixel. Pixels brighter than the selected threshold are extracted into a highlight layer. That layer is blurred outward, optionally tinted, and composited over the original image using a screen-like or additive blend. A secondary halo control can add a restrained ring around the strongest highlights. At low strength, the result resembles a polished cinematic highlight; at high strength, it can create neon or magical-light imagery.

Recommended controls

Control	Function
<code>uThreshold</code>	Decides which brightness levels are allowed to emit light.
<code>uDiffusion</code>	Controls how far and how softly the glow spreads.
<code>uIntensity</code>	Controls the brightness of the emitted glow.
<code>uColor</code>	Sets the color of the glow.
<code>uBlend</code>	Mixes the original highlight color with the selected glow color.
<code>uHalo</code>	Adds a ring-like bloom around the brightest sources.
<code>uAlpha</code>	Controls the opacity of the glow layer.

Best uses

Prism Bloom is suitable for night footage, city lights, music videos, beauty edits, gaming clips, light-based transitions, glowing titles, and footage that needs a cinematic finish. It is also useful as a subtle enhancement because it does not need to distort the entire frame.

Technical note

A high-quality version will likely use a downsampled highlight pass followed by one or more blur passes. If the first implementation must remain lightweight, a smaller blur kernel and a single-pass approximation can be used. The effect should prevent extreme intensity values from washing out the entire image.

Why Max may choose it

Prism Bloom is the strongest first contribution because its visual purpose is immediately understandable, its controls map cleanly to the existing registry schema, and it can serve both professional and social-media editing styles.

2. Mirror Orbit

Concept

Mirror Orbit converts ordinary footage into a rotating, symmetrical pattern. It reflects sections of the frame around a central point and repeats them around a circular arrangement. The result can resemble a kaleidoscope, mandala, mirrored tunnel, or psychedelic music-video background. Alight Motion's official catalog includes the underlying radial-reflection effect family.[3](#)

Exact function

The shader converts the current pixel coordinate into polar coordinates around a configurable center. The angular coordinate is divided into a chosen number of segments. Alternate segments are mirrored, then the transformed coordinates are mapped back to the source image. The segment arrangement can rotate over time, while zoom changes how much of the original image appears inside each radial section.

Recommended controls

Control	Function
<code>uSegments</code>	Sets the number of repeated radial sections.
<code>uCenter</code>	Moves the symmetry point horizontally and vertically.
<code>uRotation</code>	Rotates the repeated pattern.
<code>uZoom</code>	Controls how much of the source image is visible in each section.
<code>uMirrorMode</code>	Switches between repeated and alternating mirrored sectors.
<code>uFeather</code>	Softens seams between sections.
<code>uMix</code>	Blends the effect with the original footage.

Best uses

Mirror Orbit is excellent for music visualizers, dance edits, fashion reels, abstract backgrounds, psychedelic sequences, title cards, and transition moments. It is especially useful when Max wants the registry to include an effect that is visibly different from the current VHS, glitch, film-grain, CRT, pixelate, and duotone entries.

Technical note

The main challenges are polar-coordinate edge handling, avoiding visible seams, and keeping the center stable when the effect is animated. The shader should provide a clear edge mode so the frame does not produce unexpected transparent corners.

Why Max may choose it

Mirror Orbit is the most distinctive candidate. It can create an immediately recognizable visual identity for the registry without requiring tracking, segmentation, or external assets.

3. Velocity Veil

Concept

Velocity Veil adds softness in the direction of movement. It makes fast pans, zooms, rotations, dance movements, and animated overlays feel smoother and more like footage captured by a physical camera. Alight Motion documents motion blur derived from position, scale, and rotation animation, while Movavi promotes motion blur for making movement feel faster and more energetic.[4](#)

Exact function

The effect averages several samples taken along a motion path. When motion information is available, the sampling path follows the layer's position, scale, or rotation change. When motion vectors are unavailable, the effect can use a manually selected direction and amount as a fallback. More samples create smoother blur but increase rendering cost.

Recommended controls

Control	Function
<code>uAmount</code>	Controls the total strength of the blur.
<code>uDirection</code>	Sets the fallback direction of the blur when no motion vector is supplied.
<code>uSamples</code>	Sets the number of samples used to average the moving image.
<code>uPositionInfluence</code>	Controls how strongly positional movement contributes to blur.
<code>uScaleInfluence</code>	Controls blur caused by zooming or scaling.
<code>uRotationInfluence</code>	Controls blur caused by rotation.
<code>uMix</code>	Blends the blurred result with the original frame.

Best uses

Velocity Veil works well on action footage, sports, dance, camera pans, zoom transitions, animated stickers, title movement, and fast social edits. It should be used carefully on faces and text because excessive blur can reduce clarity.

Technical note

The cleanest implementation requires motion-vector or transform-motion input from the host application. A standalone shader can still provide directional blur, but it cannot know the true movement of individual objects in the frame. Alight Motion also notes that motion-blur behavior depends on frame rate and that motion blur can be computationally heavy.[4](#)

Why Max may choose it

Velocity Veil fills a clear gap in the existing registry and provides practical value across many editing styles. It is the best choice if the owner wants the registry to improve motion quality rather than only add stylized looks.

4. Impact Tremor

Concept

Impact Tremor produces a short, controlled burst of camera movement. It is designed for bass hits, object impacts, beat cuts, explosions, title landings, and energetic transitions. The concept

corresponds to the shake/jitter families visible in Alight Motion, CapCut transition categories, and Adobe Premiere on iPhone’s documented shake presets.[2](#) [7](#)

Exact function

The effect generates a deterministic pseudo-random offset for each frame and shapes it with an attack-and-decay envelope. The offset can affect horizontal position, vertical position, rotation, and scale. The envelope is essential: the movement should strike quickly and settle smoothly instead of behaving like continuous uncontrolled noise.

Recommended controls

Control	Function
<code>uAmplitude</code>	Sets the maximum positional displacement.
<code>uFrequency</code>	Sets how rapidly the jitter changes.
<code>uRotationAmount</code>	Adds rotational camera movement.
<code>uScaleAmount</code>	Adds a small push-in or pull-out during the tremor.
<code>uDecay</code>	Controls how quickly the effect settles after the impact.
<code>uSeed</code>	Changes the deterministic random pattern.
<code>uDirectionBias</code>	Favors horizontal, vertical, or balanced movement.
<code>uMotionBlurMix</code>	Adds a small amount of blur to integrate the shake.

Best uses

Impact Tremor is effective for music edits, sports highlights, gaming clips, action footage, beat-synced titles, comedic hits, camera knocks, and transition accents. It should be short enough to preserve readability and avoid viewer fatigue.

Technical note

If the effect is implemented purely in GLSL, it can offset UV coordinates and simulate rotation around the frame center. If the application supports layer transforms, the preferred implementation is to generate transform keyframes or a transform envelope outside the shader.

Why Max may choose it

Impact Tremor is likely the quickest of the six to make useful. It has a small control surface, a clear visual preview, and strong value for short-form content.

5. Neon Trace

Concept

Neon Trace finds the visible contours in a frame and turns them into glowing colored lines. It can create neon outlines, comic-book edges, cyberpunk line work, glowing typography, or a selective contour treatment. Alight Motion’s Edge Glow guide documents the core process: edge detection followed by controllable glow, spread, color, and threshold behavior.[8](#)

Exact function

The effect compares neighboring pixels to estimate where brightness changes rapidly. These changes form an edge map. The edge map is smoothed, expanded, colored, and blurred before being composited over the original footage. A high threshold keeps only strong contours; a low threshold reveals more detail but also makes compression noise and texture more visible.

Recommended controls

Control	Function
uThreshold	Sets the sensitivity of edge detection.
uSmoothing	Reduces noisy or overly detailed contours before glow is added.
uBlur	Controls softness and the outward reach of the glow.
uSpread	Controls the thickness of the bright contour core.
uColor	Sets the neon line color.
uGamma	Adjusts the underlying image contrast before compositing.
uInvert	Makes non-edge regions receive the glow treatment instead.
uAlpha	Controls the strength of the final contour overlay.

Best uses

Neon Trace suits music videos, gaming footage, night scenes, animated titles, fashion edits, logos, posterized portraits, and stylized backgrounds. It can also be blended lightly to emphasize a subject without completely changing the source footage.

Technical note

A Sobel or Scharr-style edge approximation is suitable for a first shader version. Smoothing should occur before the edge pass or the effect will emphasize compression artifacts. The preview image should include both a normal daylight shot and a dark subject where the neon contour is easy to see.

Why Max may choose it

Neon Trace is more visually expressive than a normal glow and more controlled than a full image-outline effect. It gives the registry a strong neon/cyber style without copying a vendor's display name.

6. Warp Surge

Concept

Warp Surge pushes the image rapidly toward or away from a focal point while adding radial streaks. It can behave as a clip effect, a transition between two shots, or a speed accent at the start or end of a scene. CapCut documents zoom and 3D-spin transition families, Alight Motion lists Zoom Blur and Zoom Streaks, and Adobe's mobile documentation includes zoom-based effects.[2](#) [7](#)

Exact function

The shader calculates the vector from the chosen focal point to the current pixel. It samples the source image at several positions along that vector, then averages the samples. The distance and direction of the samples create a zoom-streak appearance. A time or progress input can animate the amount from zero to a peak and back to zero, producing a push-through or pull-away motion.

Recommended controls

Control	Function
<code>uAmount</code>	Controls the total zoom or pull strength.
<code>uCenter</code>	Sets the focal point of the surge.
<code>uStreakLength</code>	Controls the length of the radial blur trails.
<code>uSamples</code>	Sets the number of samples used for the streak.
<code>uDirection</code>	Switches between pushing inward and pulling outward.
<code>uProgress</code>	Drives the effect through a transition or animated cycle.
<code>uEase</code>	Controls how quickly the surge starts and stops.
<code>uEdgeMode</code>	Defines whether the frame repeats, mirrors, or clamps at the edges.

Best uses

Warp Surge is useful for scene changes, travel edits, sports entrances, beat drops, title reveals, zoom-ins on faces or products, and still-image motion. Its focal point should be adjustable so the editor can direct the viewer toward the important subject.

Technical note

Radial multi-sampling can be expensive at high quality. A practical first version should expose a modest sample count and use an edge mode that prevents black gaps. The effect can later support a transition progress input supplied by the application.

Why Max may choose it

Warp Surge has the greatest value as a reusable transition. It also gives the registry a motion effect that feels more advanced than a simple scale animation while remaining achievable with a fragment shader.

Selection guide for Max

If the priority is...	Choose...	Reason
The safest and most broadly useful first contribution	Prism Bloom	Clear controls, polished result, and broad use across footage types.
A visually unique effect that distinguishes the registry	Mirror Orbit	Strong identity and no need for tracking or external masks.
Better-looking motion and camera movement	Velocity Veil	Adds realistic softness to movement and fills a current registry gap.
The fastest effect to implement	Impact Tremor	Small parameter set and immediate payoff in short-form editing.
A neon or cyberpunk signature effect	Neon Trace	Converts ordinary contours into colorful glowing lines.
A transition-focused effect	Warp Surge	Can serve as both a clip treatment and a scene transition.

Suggested implementation order

If Max wants a small but coherent first release, the recommended sequence is **Prism Bloom**, followed by **Mirror Orbit**, then **Impact Tremor**. This order begins with a polished light effect, adds a distinctive stylization effect, and finishes with an inexpensive motion effect. If the owner prioritizes transitions over standalone looks, replace Impact Tremor with Warp Surge.

The selected effect should be created in its own folder under `effects/<id>/`. The metadata must mirror the registry entry, use a semantic version such as `1.0.0`, include a declared category, list parameters that match shader uniforms, and include the compatibility object required by the repository validator. The matching registry entry should initially leave `sha256` as `{}` so the repository's CI can populate checksums after merge.[9](#)

Message for Max

Hi Max — I researched six original effect concepts for the video-effects registry. The strongest first choice is **Prism Bloom**, which creates a thresholded cinematic glow around bright areas. If you want something more distinctive, choose **Mirror Orbit**, which creates animated radial reflections. The other options are **Velocity Veil** for motion blur, **Impact Tremor** for controlled shake, **Neon Trace** for glowing outlines, and **Warp Surge** for zoom-streak transitions. Please

choose the effect you want implemented first, and I can then create the matching metadata, GLSL shader, preview image, and registry entry.

References